

SCHOLASTIC ACHIEVEMENTS

- Conferred highest grade **AP** given to **top 2%** students in Introduction to **Innovation & Entrepreneurship** course ('23)
- Secured **99.28** percentile in the Maharashtra **state-level MHT-CET** exam amongst **0.6+ million** candidates ('22)
- Recipient of the prestigious **NTSE** Scholarship offered to the top **2000** amongst **1+ million** candidates ('21)
- Selected **twice** among **top 100** applicants across India for **non-routine mathematics** training program ('17, '19)
- Scored a **perfect 100** in **Maths & Social Science** subjects in CBSE 10th board exam conducted by NCERT ('20)
- Awarded **MTSE** Scholarship offered to the **top 3** percent scorers in Std.VIII by N.Wadia College, Maharashtra ('18)

PROFESSIONAL EXPERIENCE

AI/ML Intern | *Aslan and Simba Technologies*

An e-commerce platform connecting buyers & suppliers globally for international trade

(Jun '25 - Aug '25)

- Built an e-commerce data pipeline to simplify supplier discovery, input **product queries** through **voice & images**
- Developed a **web scraping** script using **BeautifulSoup** and **JSON** to extract supplier details by product category
- Integrated **OpenAI's Whisper** for voice input **transcription** and linked it to the scraper to **automate supplier search**
- **Fine-tuned BLIP** to generate structured product descriptions from images, **enabling recognition** of unseen products

Software Engineering Intern | *KubeStream*

A platform offering AI-driven solutions for talent acquisition & cloud environment optimization

(May '25 - Jul '25)

- Developed a Talent Intelligence MVP using **React, Node.js, Python, & PostgreSQL** for a dynamic talent marketplace
- Engineered pipelines to preprocess employee & project data, generating **vector embeddings** for **semantic similarity**
- Implemented a matching engine leveraging **pgvector** for **vector search**, enabling personalized recommendations
- Built **AI agents** with **OpenAI Agents SDK** to suggest career paths, emerging opportunities, and upskilling resources

AI/ML Intern | *Samespace Labs Pvt. Ltd.*

Samespace is a comprehensive platform that streamlines and enhances business communications

(Jun '24 - Jul '24)

- Streamlined prompts by **ReAct & Reflexion**, generated **synthetic** instruction-output **data** to create dynamic questions
- Implemented **Pydantic** to extract structured output, increased the scoring accuracy by **16%** by prompt tuning
- Optimized **8-bit quantization** of **Llama3-70B** by embedding layers & **attention mechanisms**, using bitsandbytes library
- Successfully fine-tuned relevant **target modules** & hyperparameters of Llama3 & Phi3 implementing **PEFT-LoRA**

KEY PROJECTS

Music Generation & Recommendation Model | *Course Project: ME228 | Prof.S.Karagadde*

(Jan '24 - Apr '24)

- Trained a **LSTM**, a **Recurrent Neural Network** model, on Beethoven piano melodies to generate coherent sequences
- Used **AdamW** optimizer and learning rate schedulers, and dimensional reduction like **PCA, t-SNE**, to refine training
- Extracted musical notes from the corpus by **music21** and mapped the feature set to target to get a pleasing tune
- Trained a **KNN** model with song's characteristics to recommend a genre using **Euclidean** and **Manhattan** distance

Pro-Grade House Price Prediction | *Self Project*

(Apr '25)

- Designed modular ML pipeline using **ZenML** for ingesting **2.9k+** records, preprocessing, training, and deployment
- Performed **univariate/bivariate EDA** and applied statistical **imputation** on **10+** missing-value columns
- Engineered **20+** features via **log transform, Standard/MinMax scaling**; treated outliers using **IQR & Z-Score**
- Deployed Linear Regression (**$R^2 = 0.92$**) via **MLflow** with 3 workers & REST API, with **CI/CD**-compatible architecture

RAG-Based Document Retrieval | *Self Project*

(May '25)

- Parsed PDFs using **pdfplumber, PyMuPDF**, and efficiently stored chunked text with embeddings in **ChromaDB**
- Leveraged **SentenceTransformers** (all-MiniLM-L6-v2) for dense **embeddings** & HuggingFace GPT-2 for generation
- Built a **VisualRAG** class with PDF metadata, **region-based clipping**, & **Chromium Playwright** for screenshots
- Integrated **SerpAPI** for Google web search, enriching generated answers with snippets, **citations**, and **screenshots**

- Qualified for **Stage 2** among **top 4%** teams showcasing exceptional **ideation** and **innovative** problem-solving skills
- Built **AI fashion assistant** using **Groq's LLaMA-3-70B** for **multi-turn dialogue** & structured preference extraction
- Integrated **real-time** Myntra product retrieval with **streaming completions** and **context-aware link generation**
- Achieved **87% accuracy** in **matching** user style preferences during testing and **reduced response latency** by **35%**

Movie Recommender System | Self Project

(May '24)

- Developed **content-based** recommendation system using **Tf-idf Vectorizer** and **cosine similarity** for movie datasets
- Built **collaborative filtering** recommendation system using **Surprise** to predict user preferences from ratings data
- Generated **top 10** recommendations, achieved mean **RMSE** of **0.8966** and **MAE** of **0.6895** for collaborative filtering

Precipitation Prediction | WiDS, Analytics Club

(Dec '23 - Jan '24)

- Developed a ML model using **Neural Networks** and **Decision Tree classifiers** to predict the precipitation
- Used **chi-squared test** for feature selection and performed data-preprocessing & **Exploratory Data Analysis (EDA)**
- Trained a model to handle class imbalance using **SMOTE** and used **Random Forest Classifier** to train the model
- Built a **Neural Networks** model, compiled it using **Binary Crossentropy** loss function and **Adam optimizer**

CPU Scheduling Engine | Self Project

(Jun '25)

- Simulated OS-level **CPU scheduling** with **8** classic algorithms like **FCFS, RR, SRT, HRRN, Feedback, & Aging**
- Built a **flexible C++ framework** to parse user inputs, support custom quantum, and **visualize scheduling** behavior
- Developed modular logic for **preemptive** and **non-preemptive scheduling** with trace or stats output modes
- Improved **average waiting time** by up to **35%** using **Aging & FBV** over FCFS in synthetic workloads

TECHNICAL SKILLS

Languages	C, C++, Python, SQL $\text{\LaTeX}$
Libraries	NumPy, TensorFlow, PyTorch, Pandas, Matplotlib, Seaborn, NLTK, Scikit-Learn, OpenCV
Softwares	Git, GitHub, Google Colab, Jupyter Notebook, MS Excel, MATLAB, Abaqus

POSITIONS OF RESPONSIBILITY**CodeWars Mentor**

Web and Coding Club, IIT Bombay

The fourth edition of the flagship bot programming contest of Coding Club of IIT Bombay

(Mar '24)

- Responsible for **establishing communication** between the seven mentees, the club mentor, and the club managers
- Guided the mentees to **navigate** the **game engine repository** and **cleared their queries** regarding the python script

Organizer

Mood Indigo, IIT Bombay

Mood Indigo is Asia's largest college cultural festival | Footfall of 150,000+ | 230+ events

(Dec '22)

- Part of the **public organization** and crowd management committee at the 52nd edition of Mood Indigo
- Responsible for **adept direction** and **proficient management** of the crowds during a diverse array of events

KEY COURSES

Computer Science	Computer Programming and Utilisation, Logic for Computer Science, Information Retrieval & Mining for Hypertext & the Web*
Mathematics	Calculus-I, Calculus-II, Linear Algebra, Differential Equations
Machine Learning	Introduction to Machine Learning, Applied Data Science & Machine Learning, Optimization for Large Scale Machine Learning, Statistical Machine Learning & Data Mining*

(* To be completed by Nov'25)

EXTRA-CURRICULAR ACTIVITIES

- Fabricated a **model** for possible solutions of **moving sofa problem** at the 52nd AMTI Annual Conference ('17)
- Presented a **research paper** on **Combinatorics** at the 53rd Annual Conference held by AMTI ('18)
- Researched about **Criminology** as a part of **Summer of Science** by **Maths & Physics Club**, IITB ('23)
- Completed a year-long **athletics** training program in **National Sports Organization** at IITB ('23)
- Presented a **group dance** performance in the **annual dance show AIDS** by **InSync**, the dance club of IITB ('23)
- **Donated** food grains to a **NGO school** contributing towards better nutrition for **children** in rural areas ('10-'20)